Mean absolute error (days)

Median/cutoff process calibration with and without TabM residual correction · scoring dates

20/20 requested configurations complete · 2026-09-13 07:10:57 UTC

Process model
RAW · Calibration

	07	65	81	89
FL	12.0	12.9	13.9	13.6
FB	12.5	11.6	14.7	14.3
CL-FL-S	9.8	10.0	13.9	12.2
CL-FL-P	10.6	11.8	13.2	12.1
CB-FL-S	9.7	10.0	13.4	11.7
CB-FL-P	10.2	11.6	13.1	12.1
CL-FB-S	10.6	10.9	14.2	13.5
CL-FB-P	15.0	13.6	15.5	14.6
CB-FB-S	10.2	10.5	14.0	13.2
CB-FB-P	14.4	13.1	15.3	14.5

Process model
RAW · Test

	07	65	81	89
FL	11.5	12.9	13.2	15.1
FB	11.8	11.8	14.3	15.7
CL-FL-S	9.6	10.1	14.1	13.1
CL-FL-P	10.3	11.7	12.3	13.3
CB-FL-S	9.3	10.1	13.2	12.6
CB-FL-P	9.9	11.5	12.4	13.4
CL-FB-S	10.0	10.7	13.6	14.2
CL-FB-P	14.4	13.6	15.2	15.8
CB-FB-S	9.7	10.4	13.5	14.1
CB-FB-P	13.8	13.2	15.0	15.5

Process model
DEM · Calibration

	07	65	81	89
FL	11.5	11.5	12.8	13.5
FB	11.9	10.0	13.1	14.0
CL-FL-S	9.7	9.3	12.9	12.0
CL-FL-P	9.9	10.2	12.2	11.9
CB-FL-S	9.5	9.0	12.9	11.9
CB-FL-P	9.6	10.0	12.2	11.9
CL-FB-S	9.6	8.8	13.2	13.4
CL-FB-P	13.8	10.9	13.3	14.1
CB-FB-S	9.5	8.6	13.2	13.2
CB-FB-P	13.6	10.8	13.3	14.1

Process model
DEM · Test

	07	65	81	89
FL	10.8	11.7	12.6	14.2
FB	11.0	10.3	13.1	14.5
CL-FL-S	9.8	9.5	12.8	12.3
CL-FL-P	9.5	10.3	11.8	12.3
CB-FL-S	9.2	9.1	12.9	12.2
CB-FL-P	9.2	10.1	11.8	12.3
CL-FB-S	9.6	8.8	12.9	13.2
CL-FB-P	13.0	11.2	13.3	14.4
CB-FB-S	9.6	8.6	13.0	13.2
CB-FB-P	12.8	11.1	13.3	14.5

Process model + TabM residual
RAW · Calibration

	07	65	81	89
FL	4.7	4.0	5.7	7.4
FB	6.4	5.6	7.9	8.1
CL-FL-S	5.2	3.6	7.0	7.6
CL-FL-P	5.1	3.5	5.8	6.5
CB-FL-S	6.0	3.6	6.3	7.3
CB-FL-P	5.2	3.9	6.1	6.7
CL-FB-S	6.4	5.2	7.9	7.8
CL-FB-P	6.8	6.1	7.6	7.9
CB-FB-S	6.4	5.3	7.8	8.5
CB-FB-P	6.6	5.3	7.8	8.1

Process model + TabM residual
RAW · Test

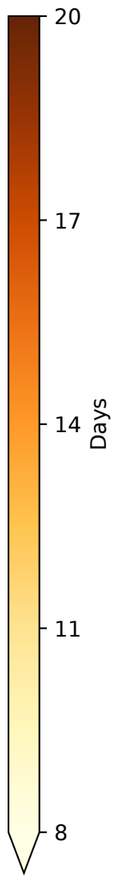
	07	65	81	89
FL	7.0	6.5	9.9	10.1
FB	7.1	6.8	10.4	10.2
CL-FL-S	6.8	6.6	11.3	10.3
CL-FL-P	6.9	6.5	10.1	10.1
CB-FL-S	7.0	6.5	10.4	9.8
CB-FL-P	6.9	6.5	9.9	9.9
CL-FB-S	7.0	6.6	10.4	10.1
CL-FB-P	7.2	6.7	10.0	10.2
CB-FB-S	6.9	6.6	10.5	10.0
CB-FB-P	7.2	6.7	10.0	9.9

Process model + TabM residual
DEM · Calibration

	07	65	81	89
FL	6.1	4.4	6.5	6.5
FB	5.3	4.6	7.0	8.5
CL-FL-S	5.5	4.2	6.9	6.9
CL-FL-P	5.4	4.4	7.0	5.8
CB-FL-S	5.0	3.9	6.9	6.8
CB-FL-P	5.0	3.8	7.0	5.5
CL-FB-S	5.6	4.3	7.0	7.5
CL-FB-P	5.7	4.7	7.5	7.9
CB-FB-S	5.6	4.3	7.2	8.2
CB-FB-P	5.1	4.9	7.3	8.8

Process model + TabM residual
DEM · Test

	07	65	81	89
FL	6.9	6.6	9.8	9.7
FB	6.9	6.6	9.9	9.9
CL-FL-S	6.8	6.4	10.4	10.3
CL-FL-P	6.8	6.5	9.9	9.8
CB-FL-S	6.7	6.5	10.4	10.3
CB-FL-P	6.8	6.5	9.8	9.7
CL-FB-S	6.8	6.5	10.7	10.7
CL-FB-P	6.9	6.4	10.1	9.9
CB-FB-S	6.7	6.4	10.6	10.6
CB-FB-P	6.9	6.5	10.0	9.9



BBCH 07 budburst · 65 flowering · 81 veraison · 89 maturity/harvest. Grey —: pending; ×: failed; ·: unrequested; n/a: unavailable in a completed unit.

MAE and bias average within sites, then weight sites equally. Bias = predicted − observed; negative is early.

TabM residual target = observed raw date − process-model raw date; features include weather through process-predicted dates.

Final hybrid dates undergo chronological correction before BBCH81/89 scoring applies the 4 November cap.

An unattained process prediction receives no ML residual; BBCH81/89 can still receive 4 November for scoring.

Calibration panels show apparent training performance, not OOF estimates. Test sites are held out. No confidence intervals.

Signed bias (days)

Median/cutoff process calibration with and without TabM residual correction · scoring dates

20/20 requested configurations complete · 2026-09-13 07:10:57 UTC

Process model
RAW · Calibration

	07	65	81	89
FL	+0.4	+0.9	+0.0	-0.1
FB	+2.2	+4.1	-4.1	+1.2
CL-FL-S	+0.7	+1.0	-0.8	+1.4
CL-FL-P	+0.9	+1.0	-0.2	+0.3
CB-FL-S	+0.9	+1.0	-0.8	+0.4
CB-FL-P	+1.0	+0.9	-0.8	-0.0
CL-FB-S	+2.1	-2.5	-3.1	+1.8
CL-FB-P	+0.7	-4.6	-6.4	+1.4
CB-FB-S	+2.0	-1.6	-2.9	+1.6
CB-FB-P	+3.5	-4.1	-5.9	+1.3

Process model
RAW · Test

	07	65	81	89
FL	+0.3	+0.5	-1.7	-1.8
FB	+2.2	+3.6	-6.1	-0.2
CL-FL-S	+0.4	+0.7	-1.2	-0.9
CL-FL-P	+0.7	+0.7	-1.8	-1.3
CB-FL-S	+0.6	+0.7	-1.6	-1.8
CB-FL-P	+0.9	+0.7	-2.3	-1.6
CL-FB-S	+2.0	-2.6	-4.8	+0.1
CL-FB-P	+0.6	-5.0	-8.4	-0.1
CB-FB-S	+1.9	-1.7	-4.5	-0.0
CB-FB-P	+3.4	-4.4	-7.9	-0.2

Process model
DEM · Calibration

	07	65	81	89
FL	+0.4	+0.6	-0.2	+0.2
FB	-0.1	+0.5	-0.6	+1.4
CL-FL-S	+0.9	+0.9	-0.9	+0.8
CL-FL-P	+1.0	+1.0	-0.2	+0.9
CB-FL-S	+0.7	+1.0	-0.5	+0.8
CB-FL-P	+1.0	+1.0	-0.2	+0.8
CL-FB-S	-0.0	+0.8	+0.2	+2.3
CL-FB-P	+1.0	+0.8	-0.3	+1.2
CB-FB-S	+0.8	+0.9	-0.4	+2.3
CB-FB-P	+1.0	+0.8	-0.3	+1.3

Process model
DEM · Test

	07	65	81	89
FL	+0.1	+0.1	-1.8	-2.2
FB	-0.4	-0.2	-2.2	-1.0
CL-FL-S	+1.0	+0.6	-1.9	-2.1
CL-FL-P	+0.6	+0.7	-1.6	-1.5
CB-FL-S	+0.4	+0.7	-1.5	-2.1
CB-FL-P	+0.7	+0.6	-1.6	-1.7
CL-FB-S	-0.0	+0.3	-1.1	-0.5
CL-FB-P	+0.5	+0.1	-1.9	-1.0
CB-FB-S	+0.9	+0.5	-1.7	-0.4
CB-FB-P	+0.5	+0.1	-1.9	-0.9

Process model + TabM residual
RAW · Calibration

	07	65	81	89
FL	-0.2	+0.5	-0.2	+0.2
FB	-0.2	+0.3	+1.3	+2.0
CL-FL-S	+1.0	+0.1	+1.2	+1.4
CL-FL-P	-0.0	-0.1	+0.5	-0.2
CB-FL-S	+0.3	+0.3	-0.2	+0.9
CB-FL-P	-0.1	+0.4	+0.3	-0.6
CL-FB-S	+1.0	+0.6	+0.6	+1.9
CL-FB-P	-0.7	+0.4	+0.3	+1.3
CB-FB-S	-0.3	-0.2	+0.0	+1.7
CB-FB-P	-0.9	+0.2	-1.1	+0.5

Process model + TabM residual
RAW · Test

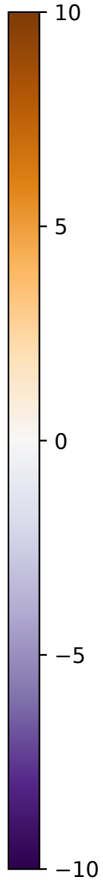
	07	65	81	89
FL	-0.6	+0.2	-1.7	-1.0
FB	-0.5	-0.2	-0.3	+0.2
CL-FL-S	+0.7	+0.1	+0.1	-0.7
CL-FL-P	-0.5	-0.3	-0.9	-1.6
CB-FL-S	-0.0	+0.1	-1.3	-1.5
CB-FL-P	-0.5	+0.1	-1.3	-1.9
CL-FB-S	+0.6	+0.2	-1.1	-0.3
CL-FB-P	-1.1	-0.1	-1.5	-0.5
CB-FB-S	-0.7	-0.5	-1.6	-0.0
CB-FB-P	-1.3	-0.2	-2.7	-1.2

Process model + TabM residual
DEM · Calibration

	07	65	81	89
FL	+0.4	+0.2	-0.1	+1.2
FB	+0.4	-0.1	+0.2	+1.0
CL-FL-S	+0.2	+0.1	+0.5	+0.9
CL-FL-P	+0.2	+0.2	+0.2	+0.3
CB-FL-S	-0.2	+0.0	+0.4	+1.5
CB-FL-P	+0.3	+0.5	+0.3	-0.4
CL-FB-S	-0.5	-0.0	+1.7	+1.3
CL-FB-P	-0.6	+1.0	+0.8	+1.3
CB-FB-S	+0.1	+0.6	+1.2	+1.7
CB-FB-P	-0.4	-0.1	+0.3	+1.2

Process model + TabM residual
DEM · Test

	07	65	81	89
FL	-0.0	-0.5	-1.5	-1.1
FB	-0.0	-0.9	-1.4	-1.6
CL-FL-S	-0.1	-0.4	-1.0	-1.8
CL-FL-P	-0.1	-0.4	-1.2	-1.9
CB-FL-S	-0.5	-0.5	-1.1	-1.5
CB-FL-P	-0.2	+0.0	-1.0	-2.7
CL-FB-S	-0.7	-0.6	+0.3	-1.9
CL-FB-P	-1.3	+0.2	-0.8	-1.1
CB-FB-S	-0.2	-0.0	-0.2	-1.2
CB-FB-P	-1.1	-0.8	-1.3	-1.2



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